Software Requirements Specification

Lylac Jewels Inventory and Order Management System

Volere Requirements Specification Template, Edition 20 — Sections 1 to 10

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Project Title: Lylac Jewels Inventory and Order Management System

Team Name: Not applicable (individual project)

Course: CINF 405: Advanced Concepts and Practices in Software Development

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Section contributions: this is an individual project. Jash Jivani wrote all sections (1–10) of this document.

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Table of Contents

1. The Purpose of the Project **4**
1a. The User Business or Background of the Project Effort 4
1b. Goals of the Project 4
2. The Stakeholders **5**
2a. The Client 5
2b. The Customer 6
2c. Other Stakeholders 7
2d. The Hands-On Users of the Product 7
2e. Personas 8
2f. Priorities Assigned to Users 9
2g. User Participation 9
2h. Maintenance Users and Service Technicians 10
3. Constraints **10**
3a. Solution Constraints 11
3b. Implementation Environment of the Current System 13
3c. Partner or Collaborative Applications 13
3d. Off-the-Shelf Software 14
3e. Anticipated Workplace Environment 14
3f. Schedule Constraints 14
3g. Budget Constraints 15
3h. Enterprise Constraints 15
4. Naming Conventions and Terminology **15**
4a. Glossary of All Terms, Including Acronyms, Used by Stakeholders Involved in the Project 15
5. Relevant Facts and Assumptions **17**
5a. Relevant Facts 17
5b. Business Rules 17
5c. Assumptions 18
6. The Scope of the Work **19**
6a. The Current Situation 19
6b. The Context of the Work 19
6c. Work Partitioning 20
6d. Specifying a Business Use Case (BUC) 21
7. Business Data Model and Data Dictionary **23**
7a. Business Data Model 23
7b. Data Dictionary 23
8. The Scope of the Product **25**
8a. Product Boundary 25
8b. Product Use Case Table 26

8c. Individual Product Use Cases....27
9. Functional Requirements....29
9a. Functional Requirements....29
10. Look and Feel Requirements....41
10a. Appearance Requirements....41
10b. Style Requirements....44

1. The Purpose of the Project

1a. The User Business or Background of the Project Effort

Lylac Jewels is a small jewelry business that my sister owns and runs. She sells pieces like necklaces, pendants, bracelets, rings and earrings. The Lylac Jewels website is still being planned and built, and right now the business does not have any software to keep track of its products, how many of each piece are in stock, or customer orders.

At the moment my sister keeps track of everything by hand, using her memory, notes and spreadsheets. That is fine when there are only a few products and orders, but it gets harder and easier to make mistakes as the business grows. When I talked with her about it, two problems stood out:

- **Overselling:** a piece can sell but the stock count does not get updated, so she might accept an order for something that is already gone.
- **Losing track of orders:** since orders are not tracked in one place, an order can be forgotten, or it is not clear if it has been confirmed, shipped or delivered yet.

I think this problem is worth solving because every oversold item or forgotten order means an unhappy customer, a refund or an apology, and wasted time for the owner. If the management system is ready before the website launches, the business will have correct product and order records from the start.

With the finished product, the owner will be able to sign in to a private dashboard, add and update products (category, description, price and quantity), see which pieces are available or sold out, record customer orders, and confirm an order only when there is enough stock, with the stock going down automatically. She will also be able to update each order’s status to pending, confirmed, shipped, delivered or canceled. If I have time, I would also like to add product images, low-stock alerts, search and filtering, and simple sales reports. Later on, the system could be connected to the full Lylac Jewels website.

This is my individual design project for CINF 405. My sister is acting as my client. She explains how the business works and gives me feedback while I build the system.

1b. Goals of the Project

Main goal: I want to give the owner of Lylac Jewels one reliable place to manage her products, stock and customer orders, so that she never accepts an order for something that is out of stock and never loses track of an order.

This mostly counts as a service goal. It helps the owner by saving her time spent checking notes and fixing mistakes, and it helps customers because they get correct answers about availability and order updates. It also gets the business ready for when the website goes live. For each goal below, I also wrote how I plan to measure whether it was met.

Goal	Advantage to the business	How success will be measured
G1. Prevent overselling	Customers are never promised a piece that is not in stock, avoiding refunds and apologies.	In acceptance testing and in the first month of use, 0 orders are confirmed when stock is insufficient, and every confirmed order reduces stock by exactly the quantities ordered.
G2. Keep every order traceable	The owner always knows what stage each order is in and can answer customer questions quickly.	100% of recorded orders show a current status, and the owner can find any order and its status in under 30 seconds.
G3. Organize the product collection	All products, prices and quantities are kept in one place instead of scattered notes.	All current products are entered in the system, and the owner can add a new product in under 2 minutes.
G4. Be easy for a non-technical owner	The owner can use the system without help, so it is actually adopted.	After a 10-minute walkthrough, the owner completes the core tasks (add product, adjust stock, record and confirm an order, update status) without help and rates the system at least 4 out of 5 for ease of use.
G5. Be ready to grow with the website	The records can later feed the public storefront without re-entering data.	The product and order data model is reviewed and accepted by the client as suitable for later connection to the website.

I will go over these goals with my client at every review meeting (see Section 2a) so the project stays on track and does not drift away from what the business actually needs.

2. The Stakeholders

2a. The Client

Client: the owner of Lylac Jewels (my sister). She owns and runs the business, and I am building the product for her. Since it is a small business she runs on her own, she is the only decision-maker. There is no bigger company or marketing department above her.

She has the final say on whether the product is acceptable. She is responsible for these decisions:

- Approving the requirements in this specification and agreeing to any later changes.
- Deciding the priority of features (for example, whether product images or low-stock alerts are added in this semester).
- Defining the business rules: product categories, what information is kept for products and orders, and how order statuses are used.
- Reviewing prototypes and the look and feel of the dashboard.
- Accepting the finished product at the final demonstration.

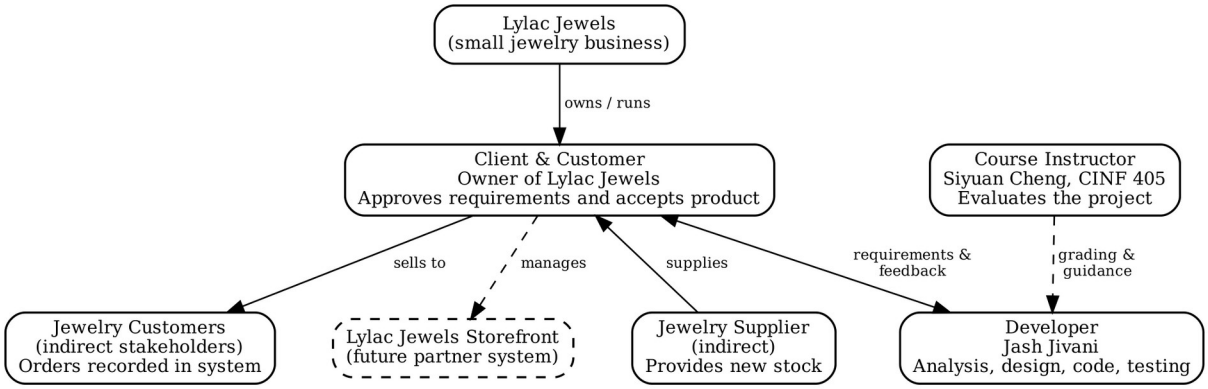


Figure 1. Where the client fits among the project's stakeholders

These are the review checkpoints I planned with my client, and what I will show her at each one:

Checkpoint	What the client receives	Client decision
1. Requirements review	This requirements document (Sections 1–10)	Approve or correct requirements and priorities
2. Database and screen design review	Data model and low-fidelity screen sketches	Approve data kept and page layout
3. Product and inventory milestone	Working login, product and stock pages	Feedback on product features
4. Order management milestone	Working order recording, confirmation and status updates	Feedback on order workflow
5. Final demonstration	Complete tested system, user notes and GitHub repository	Accept the product

2b. The Customer

Customer: the owner of Lylac Jewels. I am building the product for just one business, which is similar to in-house development, so the client and the customer are the same person. She will decide if Lylac Jewels actually starts using the system for daily work, and later if it gets connected to the website. Nobody is paying for this project, so instead of "buying" it, her decision is whether to start using it and rely on it.

The customer is responsible for these decisions:

- Whether the product is adopted for managing the real Lylac Jewels collection and orders.
- Whether real product data is entered into the system after the semester demonstration.
- Whether, and when, the system is connected to the Lylac Jewels storefront website.
- Whether optional features (images, reports, customer-facing browsing) are worth adding.

The progress indicators provided to the customer are the same as the checkpoints listed in Section 2a. At checkpoints 2, 3 and 4 she will receive a prototype she can try herself (paper sketches first, then working web pages on a laptop).

2c. Other Stakeholders

Other people and organizations who are affected by the product or whose input is needed to build it:

Stakeholder (role)	Name	Interest in the product / knowledge needed	Influence
Developer	Jash Jivani	Gathers requirements, designs, programs, tests and documents the system; maintains the GitHub repository.	High
Course instructor (evaluator)	Siyuan Cheng, CINF 405	Evaluates the requirements document and the final system against course expectations; gives guidance.	High
Domain expert	Owner of Lylac Jewels	Expert knowledge of jewelry products, pricing, stock and how orders are handled in this business.	High
Jewelry customers (members of the public)	Lylac Jewels customers	Their names, contact details and orders are stored. They benefit from accurate availability and status updates and expect their information to be kept private.	Low (indirect)
Jewelry supplier / manufacturer	Not named	Delivers new stock that the owner records in the system. Does not use the system.	Low (indirect)
Shipping carrier	Not named	Carries packages to customers; the owner records shipment and delivery in the system. Does not use the system.	Low (indirect)
Storefront website developer	Jash Jivani (future)	Will later connect the storefront website to this system, so the data must be well organized.	Medium
Usability / legal experts	None assigned	Not applicable for this semester; there is no separate usability or legal expert on this small project. Privacy of customer data is handled through the requirements.	—

2d. The Hands-On Users of the Product

The hands-on users are the people who will actually operate the product. The user characteristics are summarized below.

User role / name	Responsibilities in the work	Subject-matter experience	Technological experience	Other characteristics
Business Owner / Administrator (the owner of Lylac Jewels)	Manages the collection, stock levels and customer orders; confirms orders and updates their status.	Master: knows the products, prices, customers and business processes.	Journeyman: comfortable with email, spreadsheets, online stores and a laptop web browser; not a programmer.	Busy running the business; uses the system in short sessions, often while also answering customers. Needs clear forms and no technical terms.
Developer / Maintainer (Jash Jivani)	Sets up the system, creates the owner account, fixes errors and adds features.	Journeyman: learning the business through the client.	Journeyman: basic Python, SQL, databases and GitHub; developing web skills.	Uses the system mainly for testing and maintenance (see 2h).
Jewelry Customer (future, optional)	Would browse the available collection on a public page.	Novice.	Novice to master; general public using a phone or computer.	Not a user of the core release. Included only for the optional browsing feature.

2e. Personas

I made up these personas to help me keep the requirements focused on the kind of people who will use the product. They are fictional and do not describe real people.

Persona 1 — Maya, the business owner (primary)

Maya is 29 years old and runs a small online jewelry brand on her own. She designs the collection, photographs new pieces, answers customer messages on her phone and laptop, and packs every order herself. She lives with her family, enjoys visiting craft fairs and following fashion and design accounts on social media, and listens to pop music while she packs orders. She is organized but always short on time.

Maya is comfortable with online shopping, spreadsheets and social media, but she does not want to learn anything that feels “technical.” Her biggest worry is telling a customer that a necklace is available and then finding out it was already sold. She wants to open one page, see what is in stock, record an order in under a minute, and trust that the numbers are right. Her attitude toward new software is positive as long as it saves her time and looks as elegant as her brand.

Persona 2 — Ashley, the jewelry customer (secondary, future)

Ashley is 34, works as a nurse and lives in New Jersey. She buys jewelry online mostly as gifts for family birthdays and holidays. She shops on her phone during breaks and expects to know

right away whether an item is available and when her order will arrive. She does not use the management system; she is affected by it because the owner's answers about availability and order status come from it.

2f. Priorities Assigned to Users

User	Priority category	Reason	Importance rating (1 = low, 5 = high)
Business Owner / Administrator	Key user	The product exists to support her work; her needs decide the product's success.	5
Developer / Maintainer	Secondary user	Must be able to maintain and test the system, but the product is not built for him.	3
Jewelry Customer	Unimportant user (for this release)	Has no direct access in the core release; public browsing is optional and deferred.	1

When the needs of users conflict, the needs of the key user (the business owner) take precedence.

2g. User Participation

This table shows how much time I expect each user to spend on the project and what kind of knowledge they will give me. These are my estimates, and I will check them with my client.

User	Participation activity	Estimated time	Knowledge provided
Business Owner	Initial requirements interview (completed for the project brief) and review of this document	2 hours	Business processes, products and categories, order handling, problems with the current method
Business Owner	Design review: data kept and screen sketches	45 minutes	Which product and order details matter; preferred layout and brand look
Business Owner	Two milestone reviews of working prototypes	2 × 45 minutes	Feedback on usability, wording and missing features
Business Owner	Acceptance testing at the final demonstration	1 hour	Confirms the product meets the goals and fit criteria

User	Participation activity	Estimated time	Knowledge provided
Business Owner	Short questions by text or phone as needed	About 15 minutes per week	Clarifying business rules
Developer	All project activities	Throughout the semester	Technical design and implementation

Total estimated owner participation: about 7 to 8 hours across the semester.

2h. Maintenance Users and Service Technicians

Maintenance user: I (Jash Jivani, the developer) will be the only maintenance user during the Fall 2026 semester. That includes fixing bugs, backing up the SQLite database file, resetting the owner's password if she forgets it, adding optional features, and keeping the code up to date in the GitHub repository (https://github.com/JashJivani23/LYLAC-JEWELS).

After the semester, I plan to keep maintaining it informally since it is a family business. To make that easier, I need clear code, a README with setup steps, and a simple way to back up and restore the database.

Service technicians: Not Applicable. The product is a small web application with no special hardware, and no outside service company or IT department supports Lylac Jewels.

3. Constraints

These constraints apply to the whole product, no matter how I end up building it. I wrote each one as an atomic requirement with a rationale and a fit criterion, like the template shows. Requirement type 3 refers to this section of the template.

3a. Solution Constraints

Requirement #: C-1	**Requirement Type:** 3a	**Event/BUC/PUC #:** All
Description: The product shall be a web application whose back end is written in Python 3.12 (or later) using the Flask 3.x framework.		
Rationale: The developer has Python experience, and Flask is a lightweight framework suited to a small application that must be finished in one semester.		
Originator: Jash Jivani (developer), Project Brief		
Fit Criterion: The source code in the GitHub repository runs under Python 3.12+ with Flask 3.x listed in requirements.txt, and the application starts with a single documented command.		
Customer Satisfaction: 4	**Customer Dissatisfaction:** 4	
Priority: Must have	**Dependencies:** None	**Conflicts:** None
Supporting Materials: None		
History: Created October 16, 2026 by Jash Jivani.		

Requirement #: C-2	**Requirement Type:** 3a	**Event/BUC/PUC #:** All
Description: The product's pages shall be built with HTML5 and CSS3, using JavaScript only for optional interactive features.		
Rationale: Keeps the interface simple to build and maintain while allowing small improvements such as live search without reloading the page.		
Originator: Jash Jivani (developer), Project Brief		
Fit Criterion: Every core task (Section 8) can be completed in a current version of Chrome, Edge, Firefox or Safari, and remains possible with JavaScript turned off except for optional features.		
Customer Satisfaction: 3	**Customer Dissatisfaction:** 3	
Priority: Must have	**Dependencies:** None	**Conflicts:** None
Supporting Materials: None		
History: Created October 16, 2026 by Jash Jivani.		

Requirement #: C-3	**Requirement Type:** 3a	**Event/BUC/PUC #:** All
Description: The product shall store its data in a SQLite 3 relational database, designed so that it can be moved to MySQL later without changing the business data model.		
Rationale: SQLite needs no server setup and is suitable for one user; the owner may need a larger database later when the website goes live.		
Originator: Jash Jivani (developer), Project Brief		
Fit Criterion: All product, category, customer, order and order-item data is stored in one SQLite database file, and the schema uses only standard SQL data types and foreign keys that MySQL also supports.		
Customer Satisfaction: 3	**Customer Dissatisfaction:** 4	
Priority: Must have	**Dependencies:** None	**Conflicts:** None
Supporting Materials: None		
History: Created October 16, 2026 by Jash Jivani.		

Requirement #: C-4	**Requirement Type:** 3a	**Event/BUC/PUC #:** All
Description: All source code and project documents shall be version-controlled in the GitHub repository JashJivani23/LYLAC-JEWELS.		
Rationale: Keeps a history of changes, protects the work from loss and lets the instructor see progress.		
Originator: Course requirement; Jash Jivani		
Fit Criterion: The repository shows regular commits (at least one per active development week) and contains all code needed to run the product.		
Customer Satisfaction: 3	**Customer Dissatisfaction:** 5	
Priority: Must have	**Dependencies:** None	**Conflicts:** None
Supporting Materials: None		
History: Created October 16, 2026 by Jash Jivani.		

Requirement #: C-5	**Requirement Type:** 3a	**Event/BUC/PUC #:** All
Description: The product shall not process online payments and shall not be a mobile app.		
Rationale: Payment processing adds security and legal complexity, and mobile development is outside the semester's scope; the focus is on correct inventory and order features.		
Originator: Owner of Lylac Jewels and Jash Jivani, Project Idea		
Fit Criterion: The product contains no payment form, stores no card or bank information, and is delivered only as a web application.		
Customer Satisfaction: 3	**Customer Dissatisfaction:** 2	
Priority: Must have	**Dependencies:** None	**Conflicts:** None
Supporting Materials: None		
History: Created October 16, 2026 by Jash Jivani.		

3b. Implementation Environment of the Current System

There is no current software system for managing inventory or orders; the current work is done by hand (see Section 6a). The new product will be installed into the following environment:

- **Hardware:** the owner's personal laptop and the developer's laptop. No servers, barcode scanners or receipt printers are used.
- **Software:** a current web browser on macOS or Windows; Python 3.12+ on the computer that runs the application during the semester.
- **Network:** a home Wi-Fi internet connection. During the semester the application runs locally (localhost) for development and demonstration; hosting for real use will be chosen later.
- **Existing systems:** the Lylac Jewels storefront website is under development separately and has no connected checkout or database yet.

3c. Partner or Collaborative Applications

Lylac Jewels storefront website (future partner): the storefront that is being designed will eventually display products and could receive orders. This semester there is no runtime connection; the product must only keep its product and order data organized so that the website can use it later (see Goal G5).

GitHub: used for version control of the code. It is a development tool and does not exchange data with the running product.

No other partner applications (such as accounting, shipping or email-marketing systems) are connected to the product.

3d. Off-the-Shelf Software

The product will be built from free, open-source components rather than a purchased inventory package:

Component	Purpose	Reason for use
Flask 3.x (with Jinja2 templates)	Web framework and HTML page templates	Lightweight and fits the developer's Python skills
Werkzeug security (included with Flask)	Password hashing for the owner login	Avoids storing passwords in plain text
Flask-Login (or Flask sessions)	Keeping the owner signed in and protecting pages	Standard, well-tested approach
SQLite 3 (built into Python)	Database	No installation or server needed
pytest	Automated testing	Lets inventory and order rules be tested repeatedly

I thought about using a ready-made inventory product, but I decided not to. The owner wants a tool that fits the way she works and can connect to her own website, and for this class the point of the project is for me to build the software myself.

3e. Anticipated Workplace Environment

The owner works from home in a small space where she also photographs, packs and ships jewelry. She will use the product on a laptop, often in short sessions between other tasks, and sometimes while messaging a customer on her phone. Because of this:

- Pages must be quick to understand at a glance and let her finish a task in a few steps.
- The owner may be interrupted, so unsaved changes should not be lost without a warning, and saved data must be reliable.
- The main screen size is a laptop screen (about 13 to 15 inches). Phone use is not required this semester.
- There is no special noise, lighting or safety concern in the workplace.

3f. Schedule Constraints

The product must be completed and demonstrated within the Fall 2026 semester. Key dates follow the CINF 405 course schedule on Brightspace.

Milestone	Planned timing
Project idea, brief and introduction	Completed (September–October 2026)
Volere requirements, Sections 1–10 (this document)	October 2026

Milestone	Planned timing
Database design and owner login	Late October 2026
Product and inventory features	Early November 2026
Order management features (record, confirm, status, cancel)	Mid November 2026
Testing, interface improvements, optional features	Late November 2026
Final system, full requirements document and demonstration	December 2026 (end of semester, per course schedule)

If I start running out of time, I will drop the optional features (images, low-stock alerts, reports and customer browsing) first so that the core features still get finished.

3g. Budget Constraints

The budget for this project is $0. It is an unpaid student project, and all of the software in Section 3d is free and open source. I will use my laptop and the owner's laptop that we already have. The class demo does not need paid hosting, a domain name or any software licenses. If the owner wants to host it for real use later, she will decide on those costs then.

3h. Enterprise Constraints

Lylac Jewels is a small family business with one owner and no corporate IT policies, purchasing rules or unions. The following enterprise constraints still apply:

- The project must satisfy the CINF 405 course rules: it is completed individually, its code is kept on GitHub, and work is delivered on the course schedule.
- Customer personal information (names, emails, phone numbers and addresses) must be kept private and used only for fulfilling orders.
- Product designs and business data belong to Lylac Jewels; sample data used in class demonstrations must not expose real customers (fictional sample customers will be used).

4. Naming Conventions and Terminology

4a. Glossary of All Terms, Including Acronyms, Used by Stakeholders Involved in the Project

These are the terms used in this document so far, in alphabetical order. I expect this list to grow as the project continues.

Term	Meaning in this project
Administrator / Owner	The signed-in owner of Lylac Jewels, the only person allowed to change products, stock and orders.

Term	Meaning in this project
Available	A product whose available quantity is greater than zero.
Available Quantity (Stock)	The number of units of a product that can still be sold. It is reduced when an order is confirmed.
BUC (Business Use Case)	The business's response to a business event, described regardless of what is automated.
Canceled	Order status meaning the order will not be fulfilled.
Category	A group used to organize products, such as Necklaces, Bracelets, Rings or Earrings.
Confirmed	Order status meaning the stock check passed, the stock has been reduced, and the owner has accepted the order.
CRUD	Create, Read (Reference), Update, Delete: the four basic operations on stored data.
Customer	A person who orders jewelry from Lylac Jewels. (In Section 2b, "the customer" means the person who decides to adopt the software.)
Dashboard	The first page after sign-in, showing a summary of products and orders and links to all tasks.
Delivered	Order status meaning the package has reached the customer.
Fit Criterion	A measurable test that shows whether a requirement has been met.
Flask	The Python web framework used to build the product.
Low-Stock Threshold	A quantity set for each product; when stock is at or below it, the product is flagged as low stock (optional feature).
Order	A customer's request for one or more products, with a status.
Order Item	One line of an order: one product, the quantity ordered and the unit price at the time of the order.
Order Status	The stage of an order: Pending, Confirmed, Shipped, Delivered or Canceled.
Pending	Order status for a recorded order that has not yet been confirmed. No stock is reduced for a pending order.
Product	A jewelry item that Lylac Jewels sells, such as a particular pendant design.
PUC (Product Use Case)	The part of a business use case that is carried out by the product.
Retired Product	A product that is no longer sold but is kept because past orders refer to it.

Term	Meaning in this project
Shipped	Order status meaning the package has been handed to the shipping carrier.
SKU (Product Code)	Stock Keeping Unit: a short unique code the owner gives each product, such as LJ-NK-014.
Sold Out	A product whose available quantity is zero.
SQLite	A small relational database stored in a single file, used to store the product's data.
Storefront	The public Lylac Jewels website where customers view the collection (being developed separately).
UML	Unified Modeling Language: standard diagrams used for modeling software, such as class and use case diagrams.

5. Relevant Facts and Assumptions

5a. Relevant Facts

- Lylac Jewels is owned and operated by one person, the owner, who handles products, customer messages, packing and shipping herself.
- The business currently tracks products, stock and orders manually with notes, memory and spreadsheets; there is no inventory software.
- A storefront website for Lylac Jewels is being designed but does not yet have a connected database or checkout.
- Many jewelry pieces are made in small quantities, so the stock of a single design is often between 0 and 10 units. Selling the last unit twice is therefore a real risk.
- The business sells mainly to customers in the United States, and prices are in US dollars.
- The project is an individual student project completed by one developer in one semester.
- The developer has basic experience with Python, SQL, database design (keys, normalization) and GitHub, and is still developing web-programming skills.

5b. Business Rules

I got these business rules from my client explaining how she runs the business. I will update them as I learn more during development.

Rule #	Business rule	Reason / authority
BR-1	An order may only be confirmed if there is enough available stock for every item in the order.	Prevents overselling (Owner, Project Brief)

Rule #	Business rule	Reason / authority
BR-2	Stock is reduced when an order is confirmed, not when it is first recorded as pending.	Pending orders may still change or be canceled (Owner)
BR-3	An order is always in exactly one of five statuses: Pending, Confirmed, Shipped, Delivered or Canceled.	Owner's order process
BR-4	Orders move forward only: Pending → Confirmed → Shipped → Delivered. A Pending or Confirmed order may be Canceled; a Shipped or Delivered order may not be canceled.	Owner's order process; returns are handled outside the system
BR-5	When a Confirmed order is canceled, its quantities are added back to stock.	The items were never sent (Owner)
BR-6	Stock quantities are whole numbers and can never be negative. Prices cannot be negative.	Physical items cannot be counted below zero
BR-7	Each order item keeps the price at the time of the order, even if the product's price later changes.	Customers pay the price they were quoted
BR-8	A product that appears in any past order is retired instead of deleted, so order history stays complete.	Record keeping (Owner)
BR-9	Only the owner may add, change or delete business data.	Protects business and customer information

5c. Assumptions

#	Assumption	Effect on the project if the assumption is false
A-1	Only one person (the owner) will use the administrative dashboard; one owner account is enough.	User roles, permissions and multiple accounts would have to be added.
A-2	The owner will be available for the reviews estimated in Section 2g.	Requirements may be misunderstood; the developer would rely more on written questions and the project brief.
A-3	Orders are entered by the owner (from messages, events or the future website); customers do not place orders in the system this semester.	A customer-facing ordering page, customer accounts and stronger security would be required.
A-4	Each product belongs to one category and has a single quantity (no separate sizes or metal variants).	The data model would need product variants with their own stock.

#	Assumption	Effect on the project if the assumption is false
A-5	The number of products (under 500) and orders (under 100 per month) is small enough for SQLite on a laptop.	The product would need to move to MySQL and proper hosting sooner.
A-6	Payments, taxes, shipping fees, discounts and returns are handled outside the system.	Additional requirements and calculations would be needed; payment would raise security and legal constraints.
A-7	Fictional sample customers and representative products will be used for testing and the class demonstration.	Real customer data would need privacy protection before any demonstration.
A-8	The product will not send emails or messages to customers; the owner contacts customers herself.	Email integration would be needed, adding a partner application.

What the product will not do: process payments, calculate tax or shipping, handle returns, track packages with a carrier, send automatic customer messages, run as a mobile app, or (this semester) connect to the storefront website.

6. The Scope of the Work

6a. The Current Situation

Right now the owner does all of this work by hand. When a customer asks about or orders a piece, she checks her notes, spreadsheet or memory to see if it is in stock, replies to the customer, and writes the order down. Then she packs and ships the item and, if she remembers, updates her stock notes and the order status. Nothing connects the order to the stock count, so the two can easily end up not matching.

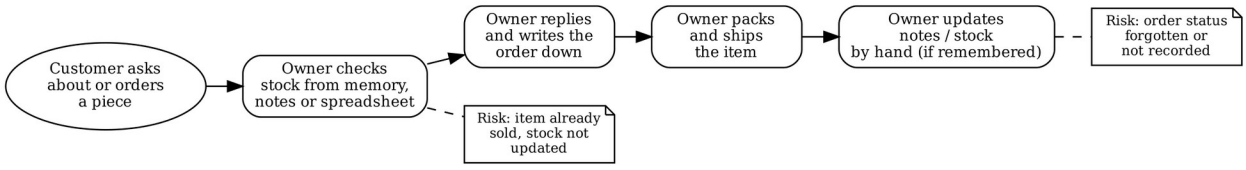


Figure 2. Current manual process and where errors occur

The weaknesses of the current situation are: (1) the stock count is updated only if the owner remembers, which causes overselling; (2) there is no single list of open orders and their status, so orders can be forgotten; (3) product details and prices are scattered across notes; and (4) there is no history to learn from, such as which pieces sell best.

6b. The Context of the Work

The work I am studying is how Lylac Jewels manages its products, inventory and customer orders. The diagram below shows the adjacent systems (the people and organizations outside the

work) and the information that goes back and forth between them and the work. The work is bigger than just the software. Things the owner does by hand, like packing orders and talking to customers, are also part of it.

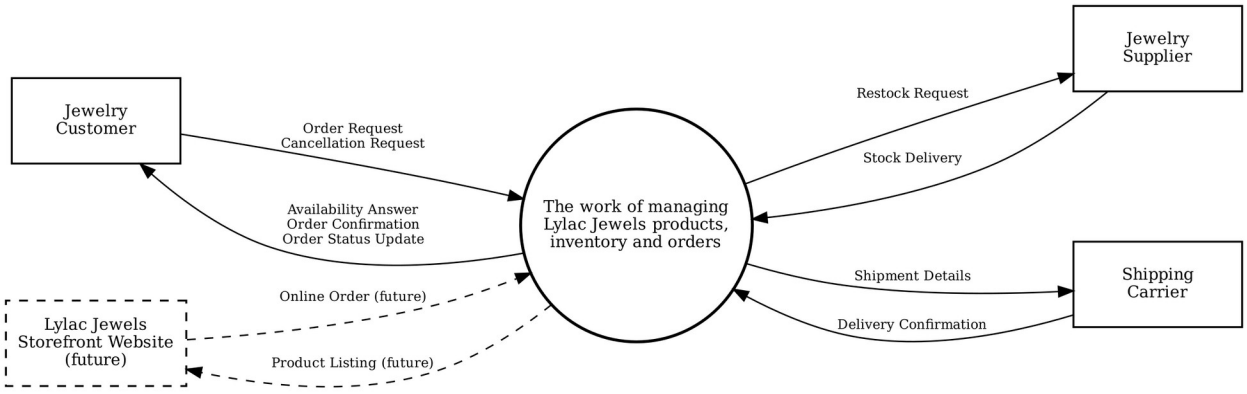


Figure 3. Work context diagram

Adjacent system	Input to the work	Output from the work
Jewelry Customer	Order Request, Cancellation Request	Availability Answer, Order Confirmation, Order Status Update
Jewelry Supplier	Stock Delivery	Restock Request
Shipping Carrier	Delivery Confirmation	Shipment Details
Lylac Jewels Storefront Website (future)	Online Order (future)	Product Listing (future)

The names of these inputs and outputs are defined in the data dictionary (Section 7b).

6c. Work Partitioning

The business events below are the happenings that make the work respond. The input and output names match the work context diagram in Section 6b.

Event Name	Input and Output	Summary of BUC
1. Owner introduces a new jewelry design	New Product Details (in)	Record the new product with its SKU, category, description, price and starting quantity.
2. Owner changes a product’s details or price	Product Change (in)	Update the product record; keep past order prices unchanged.
3. Jewelry Supplier delivers new stock	Stock Delivery (in)	Increase the product’s available quantity by the delivered amount.
4. Owner counts stock and finds a difference	Stock Correction (in)	Set the available quantity to the counted amount.

Event Name	Input and Output	Summary of BUC
5. Customer places an order	Order Request (in)	Record the customer and the requested items as a pending order.
6. Owner accepts an order	Order Confirmation (out), Availability Answer (out)	Check stock for every item; if enough, reduce stock and confirm the order; if not, keep it pending and tell the customer what is unavailable.
7. Owner ships an order	Shipment Details (out), Order Status Update (out)	Record the order as shipped and inform the customer.
8. Shipping Carrier delivers the package	Delivery Confirmation (in), Order Status Update (out)	Record the order as delivered.
9. Customer cancels an order	Cancellation Request (in), Order Status Update (out)	Mark the order canceled; if it was confirmed, return its quantities to stock.
10. Owner decides a product will no longer be sold	Product Retirement (in)	Retire the product so it cannot be ordered, keeping it in past orders.
11. Time to review stock levels (weekly)	Low-Stock List (out), Restock Request (out)	List products at or below their low-stock threshold so the owner can reorder (optional).
12. Owner needs to know how sales are going (monthly)	Sales Summary (out)	Summarize orders and units sold for a period (optional).

6d. Specifying a Business Use Case (BUC)

BUC-6: Accept a customer's order (business event 6; BUC numbers follow event numbers)

Item	Specification
Trigger	An Order Request arrives from a Jewelry Customer and the owner decides to accept it.
Preconditions	The products requested exist in the collection and are not retired.
Interested stakeholders	Owner, Jewelry Customer
Active stakeholder	Owner

Item	Specification
Normal case	1. Record the customer’s name and contact details. 2. Record each requested product and quantity as a pending order. 3. Check the available quantity of every requested product. 4. If every product has enough stock, reduce the stock of each product by its ordered quantity and mark the order Confirmed. 5. Send the Order Confirmation to the customer. 6. Later, pack the order and hand Shipment Details to the Shipping Carrier.
Alternative case	4a. If any product does not have enough stock, keep the order Pending, do not change any stock, and send an Availability Answer telling the customer which piece is unavailable. The customer may change the order or cancel it.
Exception	4b. If the customer cancels before shipping, follow business event 9.
Outcome	The order is confirmed and stock is accurate, or the customer is told what is unavailable before any stock is promised.
Business fit criterion	Every confirmed order has enough stock for all its items, and the stock after confirmation equals the stock before minus the quantities ordered.

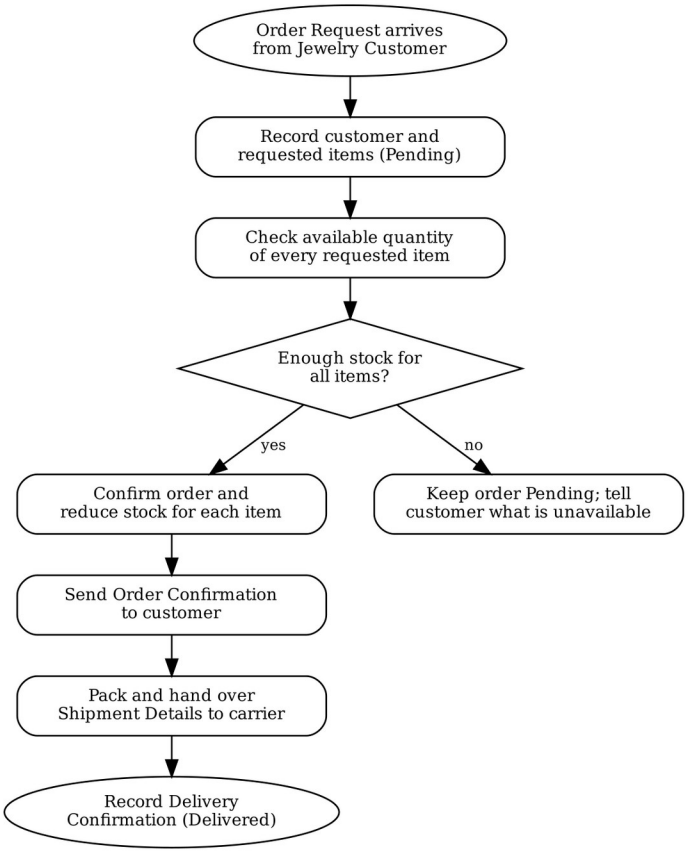


Figure 4. Activity diagram for BUC-6, Accept a customer’s order

7. Business Data Model and Data Dictionary

7a. Business Data Model

Below is my first version of the business data model, drawn as a UML class model. It shows the main things the owner needs to keep track of for her collection and her orders. This is a business model, not the final database design, so I left out design details like the owner's login account.

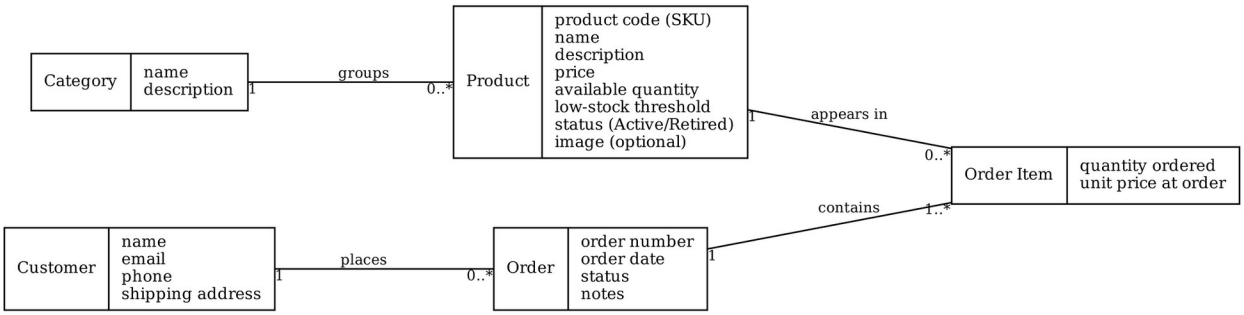


Figure 5. Business data model (UML class model)

Class	Relationships	Attributes
Category	Groups 0..* Products	name, description
Product	Belongs to 1 Category; appears in 0..* Order Items	product code (SKU), name, description, price, available quantity, low-stock threshold, status (Active/Retired), image (optional)
Customer	Places 0..* Orders	name, email, phone, shipping address
Order	Placed by 1 Customer; contains 1..* Order Items	order number, order date, status, notes
Order Item	Belongs to 1 Order; refers to 1 Product	quantity ordered, unit price at order

7b. Data Dictionary

The data dictionary defines the classes, attributes and data flows used in this specification. The same terms are used in the requirements in Sections 9 and 10. Notation: "+" means "and"; "{ }" means "one or more"; "( )" means optional.

Name	Content	Type
Category	Category Name + (Category Description)	Class
Customer	Customer Name + Customer Email + (Customer Phone) + Shipping Address	Class

Name	Content	Type
Order	Order Number + Order Date + Order Status + (Order Notes) + Customer + {Order Item}	Class
Order Item	Product + Quantity Ordered + Unit Price at Order	Class
Product	SKU + Product Name + Category + Product Description + Price + Available Quantity + (Low-Stock Threshold) + Product Status + (Product Image)	Class
Availability Answer	Order Number + {Product Name + Available Quantity}	Dataflow
Cancellation Request	Order Number + Customer Name	Dataflow
Delivery Confirmation	Order Number + Delivery Date	Dataflow
Low-Stock List	{SKU + Product Name + Available Quantity + Low-Stock Threshold}	Dataflow
New Product Details	SKU + Product Name + Category + Product Description + Price + Available Quantity	Dataflow
Online Order (future)	Customer + {Product + Quantity Ordered}	Dataflow
Order Confirmation	Order Number + Order Date + {Product Name + Quantity Ordered + Unit Price at Order} + Order Total	Dataflow
Order Request	Customer + {Product + Quantity Ordered}	Dataflow
Order Status Update	Order Number + Order Status	Dataflow
Product Change	SKU + (Product Name) + (Category) + (Product Description) + (Price)	Dataflow
Product Listing (future)	{Product Name + Product Description + Price + Availability + (Product Image)}	Dataflow
Product Retirement	SKU	Dataflow
Restock Request	{Product Name + Quantity Needed}	Dataflow
Sales Summary	Period + Number of Orders + {Product Name + Units Sold} + Sales Total	Dataflow
Shipment Details	Order Number + Customer Name + Shipping Address + Ship Date	Dataflow
Stock Correction	SKU + Counted Quantity	Dataflow
Stock Delivery	SKU + Quantity Received	Dataflow

Name	Content	Type
Availability	*“Available” if Available Quantity > 0; “Sold Out” if Available Quantity = 0*	Attribute/ Element
Available Quantity	*Whole number of units, 0 or more*	Attribute/ Element
Customer Email	*A valid email address, for example name@example.com*	Attribute/ Element
Low-Stock Threshold	*Whole number, 0 or more; default 2*	Attribute/ Element
Order Date	*Date and time the order was recorded*	Attribute/ Element
Order Number	*Unique number given by the product, for example 1001*	Attribute/ Element
Order Status	*One of: Pending, Confirmed, Shipped, Delivered, Canceled*	Attribute/ Element
Order Total	*Sum of (Quantity Ordered × Unit Price at Order) for all Order Items, in US dollars*	Attribute/ Element
Price / Unit Price at Order	*US dollars and cents, 0.00 or more*	Attribute/ Element
Product Status	*Active or Retired*	Attribute/ Element
Quantity Ordered	*Whole number, 1 or more*	Attribute/ Element
SKU	*Unique product code chosen by the owner, up to 20 letters, numbers or dashes, for example LJ-NK-014*	Attribute/ Element

8. The Scope of the Product

8a. Product Boundary

The product use case diagram shows which parts of the work the product will handle. The owner is the only actor in the main version. The dashed use cases are optional features that I will only build if I have time. The Jewelry Customer, Supplier and Carrier stay outside the product, because the owner talks to them herself and does the packing and shipping by hand.

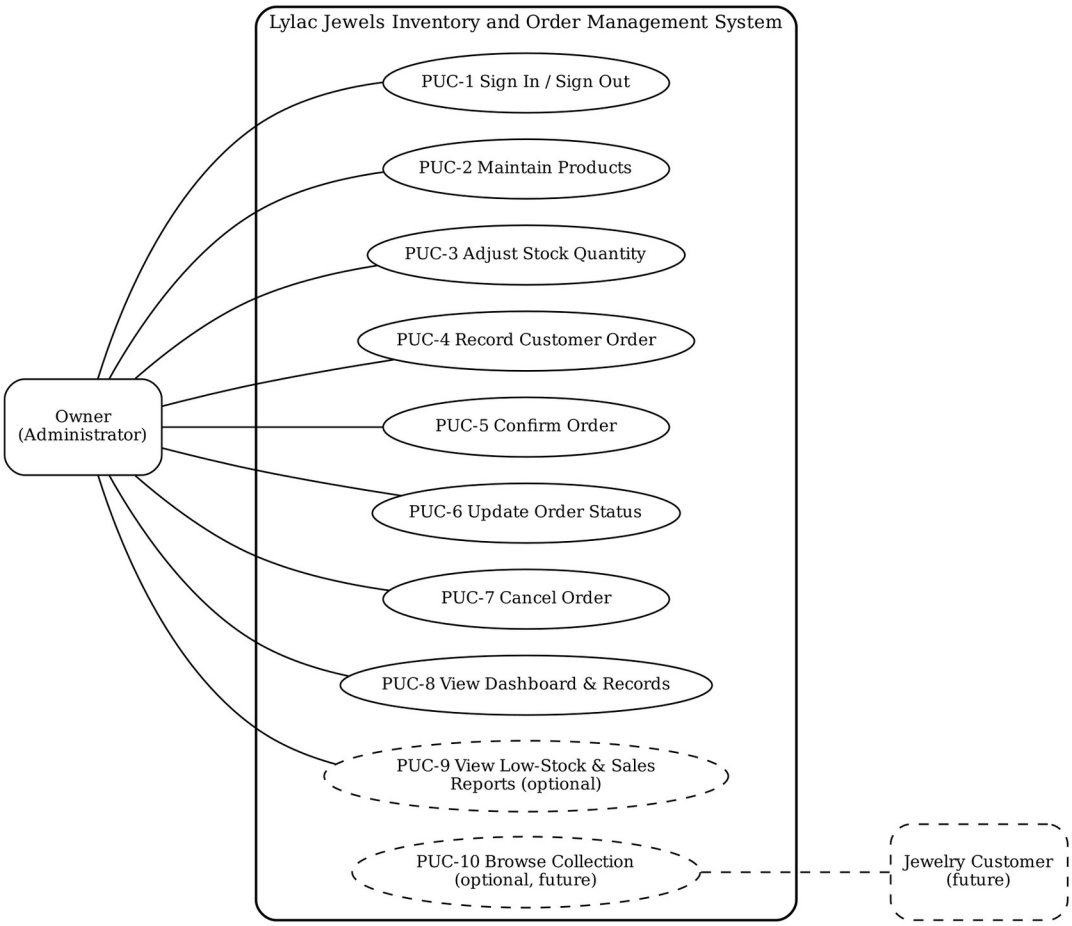


Figure 6. Product use case diagram (product boundary)

Inside the product boundary: owner sign-in, product and category records, stock quantities, customer and order records, the stock check and stock reduction at confirmation, order status changes, cancellation with stock restoration, and the dashboard. **Outside the boundary:** payments, packing and shipping, carrier tracking, customer messaging, returns, and the public storefront.

8b. Product Use Case Table

PUC No	PUC Name	Actors/Users	Input & Output
1	Sign In / Sign Out (all BUCs)	Owner	Username and Password (in); Dashboard (out)
2	Maintain Products (BUC 1, 2, 10)	Owner	New Product Details, Product Change, Product Retirement (in); Product record (out)
3	Adjust Stock Quantity (BUC 3, 4)	Owner	Stock Delivery, Stock Correction (in); Available Quantity and Availability (out)

PUC No	PUC Name	Actors/Users	Input & Output
4	Record Customer Order (BUC 5)	Owner	Order Request (in); Pending Order (out)
5	Confirm Order (BUC 6)	Owner	Pending Order (in); Order Confirmation or list of unavailable items (out)
6	Update Order Status (BUC 7, 8)	Owner	Shipment Details, Delivery Confirmation (in); Order Status Update (out)
7	Cancel Order (BUC 9)	Owner	Cancellation Request (in); Order Status Update, restored stock (out)
8	View Dashboard & Records (all BUCs)	Owner	Search or filter choice (in); Product list, Order list, order details, summary counts (out)
9	View Low-Stock & Sales Reports (BUC 11, 12) — optional	Owner	Report period (in); Low-Stock List, Sales Summary (out)
10	Browse Collection — optional, future	Jewelry Customer	Category choice (in); Product Listing (out)

8c. Individual Product Use Cases

Every product use case except PUC-1 and PUC-10 requires the owner to be signed in. If the owner enters invalid data, the product saves nothing, keeps what she typed, and explains what to fix.

PUC-1: Sign In / Sign Out

The owner opens the system and sees the sign-in page. She enters her username and password. If they are correct, the dashboard opens; if not, the product shows “Incorrect username or password” and does not reveal which one was wrong. Any attempt to open a protected page without signing in returns her to the sign-in page. When she selects Sign Out, the session ends.

PUC-2: Maintain Products

The owner selects Add Product, enters the SKU, name, category, description, price and starting quantity, and saves. The product checks that required fields are filled in, the SKU is unique, the price is not negative and the quantity is a whole number of 0 or more. To edit, she opens a product, changes details and saves. To remove a product, she selects Delete: a product that has never been ordered is deleted after confirmation; a product that appears in past orders is retired instead (BR-8).

PUC-3: Adjust Stock Quantity

The owner opens a product and either adds a received quantity (Stock Delivery) or enters the counted quantity (Stock Correction). The product validates the number, saves the new available quantity and immediately shows “Available” or “Sold Out.”

PUC-4: Record Customer Order

The owner selects New Order, chooses an existing customer or enters a new one, then adds one or more active products with quantities. The product shows each item’s current price and the order total, saves the order as Pending with the price at the time of the order, and does not change stock (BR-2). An empty order or a quantity below 1 is rejected.

PUC-5: Confirm Order

I think this is the most important use case, because it is where orders and inventory connect. Figure 7 shows the scenario.

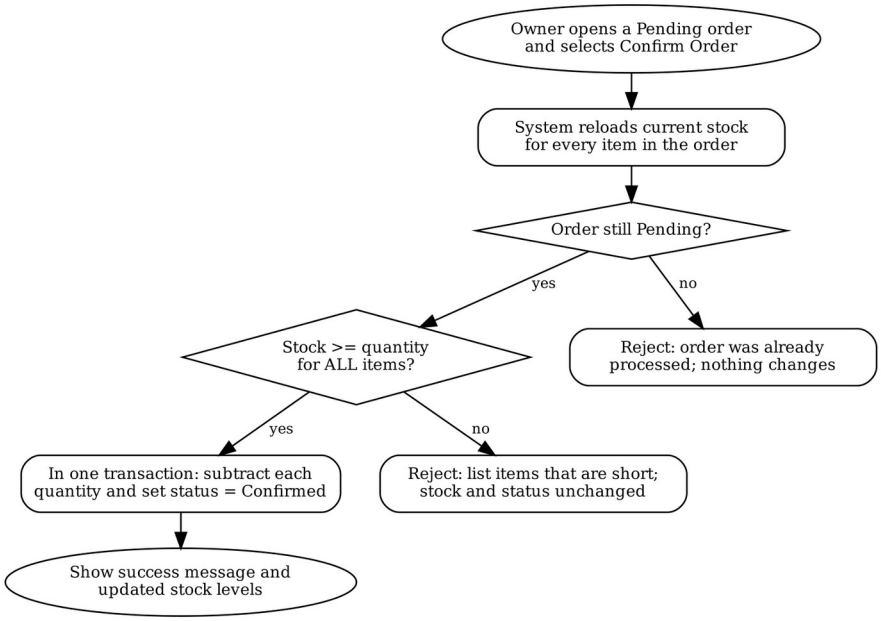


Figure 7. Activity diagram for PUC-5, Confirm Order

- **Normal case:** the owner opens a Pending order and selects Confirm Order. The product reloads the current stock for each item, checks that every item has enough stock, then in a single transaction subtracts each quantity and sets the status to Confirmed. It shows the new stock levels.
- **Alternative case (insufficient stock):** if any item is short, nothing is changed and the product lists each short item with its available quantity, so the owner can tell the customer.
- **Exception (already processed):** if the order is no longer Pending (for example, the confirm button was pressed twice), the product refuses the request so stock is never reduced twice.
- **Fit criterion:** with stock Necklace A = 5 and Bracelet B = 2, confirming an order for 2 A and 1 B leaves A = 3, B = 1 and the order Confirmed; an order for 3 B is rejected and leaves all stock unchanged.

PUC-6: Update Order Status and PUC-7: Cancel Order

The owner opens an order and chooses the next allowed status. The allowed changes are shown in Figure 8. Canceling a Pending order changes no stock; canceling a Confirmed order returns its quantities to stock once. Shipped and Delivered orders cannot be canceled.

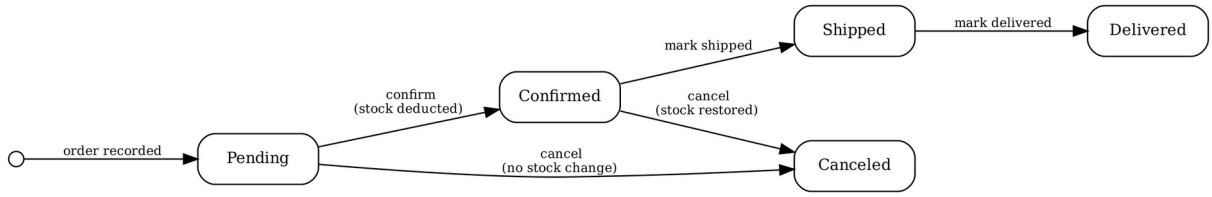


Figure 8. Order status state diagram for PUC-6 and PUC-7

PUC-8: View Dashboard & Records

After sign-in, the dashboard shows the number of Pending orders, Confirmed orders waiting to ship, sold-out products and (if built) low-stock products, with links to each list. The owner can open the product list or order list, search by name or SKU, filter by category or status, and open any record to see its details. If a list is empty, a friendly message explains how to add the first record.

PUC-9: View Low-Stock & Sales Reports (optional)

The owner opens Reports and sees products at or below their low-stock threshold, and a summary of orders and units sold for a chosen month. Canceled orders are excluded from sales totals.

PUC-10: Browse Collection (optional, future)

A public page would list active products with name, price, picture and availability, without any prices or quantities that the owner marks as private. This use case is deferred until the core features are complete.

9. Functional Requirements

9a. Functional Requirements

I wrote each functional requirement using the Volere atomic requirement shell. Requirement type 9 means a functional requirement. Customer satisfaction (1 = uninterested, 5 = extremely pleased if it is built) and customer dissatisfaction (1 = hardly matters, 5 = extremely displeased if it is left out) are my first estimates based on talking with my client, and I will confirm them with her at our next review. For priority, I used Must have, Should have and Could have.

Requirement #: FR-1	**Requirement Type:** 9	**Event/BUC/PUC #:** PUC-1
Description: The product shall allow the owner to sign in with a username and password before any administrative page is shown.		
Rationale: Business and customer data must be protected from anyone other than the owner (BR-9).		
Originator: Owner of Lylac Jewels (client)		
Fit Criterion: Entering the correct username and password opens the dashboard; an incorrect password shows an error and opens no administrative page; passwords are stored only as hashes in the database.		
Customer Satisfaction: 4	**Customer Dissatisfaction:** 5	
Priority: Must have	**Dependencies:** None	**Conflicts:** None
Supporting Materials: None		
History: Created October 16, 2026 by Jash Jivani.		

Requirement #: FR-2	**Requirement Type:** 9	**Event/BUC/PUC #:** PUC-1
Description: The product shall return any request for an administrative page from someone who is not signed in to the sign-in page.		
Rationale: Typing a page address directly must not bypass the login.		
Originator: Jash Jivani (developer)		
Fit Criterion: Requesting each administrative page address while signed out shows the sign-in page and displays or changes no business data.		
Customer Satisfaction: 3	**Customer Dissatisfaction:** 5	
Priority: Must have	**Dependencies:** FR-1	**Conflicts:** None
Supporting Materials: None		
History: Created October 16, 2026 by Jash Jivani.		

Requirement #: FR-3	**Requirement Type:** 9	**Event/BUC/PUC #:** PUC-1
Description: The product shall allow the owner to sign out, ending her session.		
Rationale: Prevents someone else using the laptop from accessing the dashboard.		
Originator: Jash Jivani (developer)		
Fit Criterion: After signing out, pressing the browser's Back button or reopening a protected page shows the sign-in page.		
Customer Satisfaction: 2	**Customer Dissatisfaction:** 4	
Priority: Must have	**Dependencies:** FR-1	**Conflicts:** None
Supporting Materials: None		
History: Created October 16, 2026 by Jash Jivani.		

Requirement #: FR-4	**Requirement Type:** 9	**Event/BUC/PUC #:** BUC 1 / PUC-2
Description: The product shall allow the owner to add a product with its SKU, name, category, description, price and available quantity.		
Rationale: The collection must be recorded in one place (Goal G3).		
Originator: Owner of Lylac Jewels (client)		
Fit Criterion: A product saved with valid details appears in the product list, and reopening it shows exactly the values entered. Adding a product takes the owner less than 2 minutes.		
Customer Satisfaction: 5	**Customer Dissatisfaction:** 5	
Priority: Must have	**Dependencies:** FR-1	**Conflicts:** None
Supporting Materials: None		
History: Created October 16, 2026 by Jash Jivani.		

Requirement #: FR-5	**Requirement Type:** 9	**Event/BUC/PUC #:** BUC 1, 2 / PUC-2
Description: The product shall reject a product that is missing a required field, has a duplicate SKU, a negative price, or a quantity that is not a whole number of 0 or more.		
Rationale: Invalid product data would make stock and prices unreliable (BR-6).		
Originator: Jash Jivani (developer)		
Fit Criterion: Each invalid case is refused with a message naming the field to fix, no record is created or changed, and the values the owner typed remain in the form.		
Customer Satisfaction: 3	**Customer Dissatisfaction:** 4	
Priority: Must have	**Dependencies:** FR-4	**Conflicts:** None
Supporting Materials: None		
History: Created October 16, 2026 by Jash Jivani.		

Requirement #: FR-6	**Requirement Type:** 9	**Event/BUC/PUC #:** BUC 2 / PUC-2
Description: The product shall allow the owner to change a product’s name, category, description and price.		
Rationale: Products and prices change over time.		
Originator: Owner of Lylac Jewels (client)		
Fit Criterion: After a price change, the product list shows the new price, and existing orders still show the price recorded when they were placed (BR-7).		
Customer Satisfaction: 4	**Customer Dissatisfaction:** 4	
Priority: Must have	**Dependencies:** FR-4	**Conflicts:** None
Supporting Materials: None		
History: Created October 16, 2026 by Jash Jivani.		

Requirement #: FR-7	**Requirement Type:** 9	**Event/BUC/PUC #:** BUC 10 / PUC-2
Description: The product shall delete a product that has never been ordered, and retire (instead of delete) a product that appears in any order.		
Rationale: Removing old products must not break order history (BR-8).		
Originator: Owner of Lylac Jewels (client)		
Fit Criterion: An unused product disappears after the owner confirms deletion. A product used in an order is marked Retired, still appears in that order, and can no longer be added to new orders.		
Customer Satisfaction: 3	**Customer Dissatisfaction:** 4	
Priority: Must have	**Dependencies:** FR-4, FR-12	**Conflicts:** None
Supporting Materials: None		
History: Created October 16, 2026 by Jash Jivani.		

Requirement #: FR-8	**Requirement Type:** 9	**Event/BUC/PUC #:** BUC 3, 4 / PUC-3
Description: The product shall allow the owner to add received stock to a product or set its available quantity to a counted amount.		
Rationale: The stock count must match the real pieces on hand.		
Originator: Owner of Lylac Jewels (client)		
Fit Criterion: Adding 3 to a quantity of 4 saves 7; setting a counted amount of 0 saves 0; a negative or fractional amount is rejected and the saved quantity is unchanged.		
Customer Satisfaction: 5	**Customer Dissatisfaction:** 5	
Priority: Must have	**Dependencies:** FR-4	**Conflicts:** None
Supporting Materials: None		
History: Created October 16, 2026 by Jash Jivani.		

Requirement #: FR-9	**Requirement Type:** 9	**Event/BUC/PUC #:** PUC-3, PUC-8
Description: The product shall show each product as “Available” when its available quantity is above zero and “Sold Out” when it is zero.		
Rationale: The owner needs to see at a glance what can be sold.		
Originator: Owner of Lylac Jewels (client)		
Fit Criterion: A product with quantity 1 shows “Available”; after its last unit is confirmed in an order it shows “Sold Out” without any manual change.		
Customer Satisfaction: 4	**Customer Dissatisfaction:** 4	
Priority: Must have	**Dependencies:** FR-8, FR-14	**Conflicts:** None
Supporting Materials: None		
History: Created October 16, 2026 by Jash Jivani.		

Requirement #: FR-10	**Requirement Type:** 9	**Event/BUC/PUC #:** BUC 5 / PUC-4
Description: The product shall allow the owner to record a customer’s name, email, phone number and shipping address, or select an existing customer.		
Rationale: Orders must be linked to the person who placed them.		
Originator: Owner of Lylac Jewels (client)		
Fit Criterion: A new customer saved with an order can be selected for a later order without retyping, and a customer record without a name or valid email is rejected.		
Customer Satisfaction: 4	**Customer Dissatisfaction:** 4	
Priority: Must have	**Dependencies:** FR-1	**Conflicts:** None
Supporting Materials: None		
History: Created October 16, 2026 by Jash Jivani.		

Requirement #: FR-11	**Requirement Type:** 9	**Event/BUC/PUC #:** BUC 5 / PUC-4
Description: The product shall allow the owner to record an order containing one or more active products and a quantity of at least 1 for each.		
Rationale: Customer requests need to be captured before they are confirmed.		
Originator: Owner of Lylac Jewels (client)		
Fit Criterion: An order with two products is saved and reopened with the same customer, products and quantities; an order with no items or a quantity of 0 is rejected; retired products cannot be selected.		
Customer Satisfaction: 5	**Customer Dissatisfaction:** 5	
Priority: Must have	**Dependencies:** FR-4, FR-10	**Conflicts:** None
Supporting Materials: None		
History: Created October 16, 2026 by Jash Jivani.		

Requirement #: FR-12	**Requirement Type:** 9	**Event/BUC/PUC #:** BUC 5 / PUC-4
Description: The product shall save each new order with the status Pending, the date recorded, a unique order number and each item’s price at the time of the order, without changing stock.		
Rationale: Stock is only committed at confirmation (BR-2, BR-7).		
Originator: Owner of Lylac Jewels (client)		
Fit Criterion: Immediately after saving a new order, its status is Pending, it has a new order number, and every product’s available quantity is the same as before.		
Customer Satisfaction: 4	**Customer Dissatisfaction:** 5	
Priority: Must have	**Dependencies:** FR-11	**Conflicts:** None
Supporting Materials: None		
History: Created October 16, 2026 by Jash Jivani.		

Requirement #: FR-13	**Requirement Type:** 9	**Event/BUC/PUC #:** PUC-4, PUC-8
Description: The product shall calculate and display the order total as the sum of quantity times unit price for all items.		
Rationale: The owner needs to tell the customer the amount owed.		
Originator: Owner of Lylac Jewels (client)		
Fit Criterion: An order for 2 items at $45.00 and 1 item at $30.50 shows a total of $120.50.		
Customer Satisfaction: 4	**Customer Dissatisfaction:** 3	
Priority: Should have	**Dependencies:** FR-12	**Conflicts:** None
Supporting Materials: None		
History: Created October 16, 2026 by Jash Jivani.		

Requirement #: FR-14	**Requirement Type:** 9	**Event/BUC/PUC #:** BUC 6 / PUC-5
Description: The product shall confirm a Pending order only when every item has enough available stock, reducing the available quantity of each product by its ordered quantity in the same single operation.		
Rationale: This is the main reason for the project: prevent overselling (BR-1, Goal G1).		
Originator: Owner of Lylac Jewels (client)		
Fit Criterion: With stock A = 5 and B = 2, confirming 2 A and 1 B results in A = 3, B = 1 and status Confirmed. If the save fails part way, no quantity is changed.		
Customer Satisfaction: 5	**Customer Dissatisfaction:** 5	
Priority: Must have	**Dependencies:** FR-8, FR-12	**Conflicts:** None
Supporting Materials: Section 8c, Figure 7		
History: Created October 16, 2026 by Jash Jivani.		

Requirement #: FR-15	**Requirement Type:** 9	**Event/BUC/PUC #:** BUC 6 / PUC-5
Description: The product shall refuse to confirm an order when any item has insufficient stock, list each short item with its available quantity, and leave all stock and the order status unchanged.		
Rationale: The owner must know exactly what to tell the customer.		
Originator: Owner of Lylac Jewels (client)		
Fit Criterion: With B = 2, an order for 3 B is refused, the message names Bracelet B with 2 available, and both stock and the Pending status are unchanged.		
Customer Satisfaction: 4	**Customer Dissatisfaction:** 5	
Priority: Must have	**Dependencies:** FR-14	**Conflicts:** None
Supporting Materials: None		
History: Created October 16, 2026 by Jash Jivani.		

Requirement #: FR-16	**Requirement Type:** 9	**Event/BUC/PUC #:** BUC 6 / PUC-5
Description: The product shall refuse to confirm an order that is not Pending.		
Rationale: Pressing Confirm twice must never reduce stock twice.		
Originator: Jash Jivani (developer)		
Fit Criterion: Submitting confirmation a second time for the same order changes no stock and shows that the order was already confirmed.		
Customer Satisfaction: 3	**Customer Dissatisfaction:** 5	
Priority: Must have	**Dependencies:** FR-14	**Conflicts:** None
Supporting Materials: None		
History: Created October 16, 2026 by Jash Jivani.		

Requirement #: FR-17	**Requirement Type:** 9	**Event/BUC/PUC #:** BUC 7, 8 / PUC-6
Description: The product shall allow the owner to change an order's status only along the allowed path Confirmed → Shipped → Delivered.		
Rationale: Keeps order tracking consistent (BR-3, BR-4, Goal G2).		
Originator: Owner of Lylac Jewels (client)		
Fit Criterion: A Confirmed order can be marked Shipped and then Delivered, each change is saved with its date, and changing Pending directly to Shipped or Delivered is refused.		
Customer Satisfaction: 4	**Customer Dissatisfaction:** 5	
Priority: Must have	**Dependencies:** FR-14	**Conflicts:** None
Supporting Materials: Figure 8		
History: Created October 16, 2026 by Jash Jivani.		

Requirement #: FR-18	**Requirement Type:** 9	**Event/BUC/PUC #:** BUC 9 / PUC-7
Description: The product shall allow the owner to cancel a Pending or Confirmed order, returning the quantities of a Confirmed order to stock exactly once.		
Rationale: Items from a canceled order can be sold again (BR-5).		
Originator: Owner of Lylac Jewels (client)		
Fit Criterion: Canceling a Pending order changes no stock. Confirming 2 units from a stock of 5 and then canceling restores the stock to 5. Shipped and Delivered orders cannot be canceled.		
Customer Satisfaction: 4	**Customer Dissatisfaction:** 5	
Priority: Must have	**Dependencies:** FR-14, FR-17	**Conflicts:** None
Supporting Materials: None		
History: Created October 16, 2026 by Jash Jivani.		

Requirement #: FR-19	**Requirement Type:** 9	**Event/BUC/PUC #:** PUC-8
Description: The product shall display lists of products and orders that the owner can search by name, SKU or order number and filter by category or order status.		
Rationale: The owner must find any record quickly (Goal G2).		
Originator: Owner of Lylac Jewels (client)		
Fit Criterion: With 50 sample products and 30 sample orders, the owner finds a named order and reads its status in under 30 seconds.		
Customer Satisfaction: 4	**Customer Dissatisfaction:** 3	
Priority: Should have	**Dependencies:** FR-4, FR-12	**Conflicts:** None
Supporting Materials: None		
History: Created October 16, 2026 by Jash Jivani.		

Requirement #: FR-20	**Requirement Type:** 9	**Event/BUC/PUC #:** PUC-8
Description: The product shall show a dashboard after sign-in with counts of Pending orders, Confirmed orders not yet shipped, and Sold Out products.		
Rationale: Gives the owner a quick picture of what needs attention.		
Originator: Owner of Lylac Jewels (client)		
Fit Criterion: Each count on the dashboard equals the number of matching records in the lists after any change.		
Customer Satisfaction: 4	**Customer Dissatisfaction:** 3	
Priority: Should have	**Dependencies:** FR-9, FR-12	**Conflicts:** None
Supporting Materials: None		
History: Created October 16, 2026 by Jash Jivani.		

Requirement #: FR-21	**Requirement Type:** 9	**Event/BUC/PUC #:** BUC 11 / PUC-9
Description: The product shall list the products whose available quantity is at or below their low-stock threshold.		
Rationale: Helps the owner reorder before a piece sells out.		
Originator: Owner of Lylac Jewels (client)		
Fit Criterion: With a threshold of 2, products with quantity 0, 1 or 2 appear in the list and a product with quantity 3 does not.		
Customer Satisfaction: 4	**Customer Dissatisfaction:** 2	
Priority: Could have	**Dependencies:** FR-8	**Conflicts:** None
Supporting Materials: None		
History: Created October 16, 2026 by Jash Jivani.		

Requirement #: FR-22	**Requirement Type:** 9	**Event/BUC/PUC #:** BUC 12 / PUC-9
Description: The product shall summarize, for a chosen month, the number of orders, units sold per product and total sales, excluding canceled orders.		
Rationale: Shows the owner which pieces sell best.		
Originator: Owner of Lylac Jewels (client)		
Fit Criterion: For a test month with known sample orders, the report totals match a hand calculation exactly.		
Customer Satisfaction: 3	**Customer Dissatisfaction:** 1	
Priority: Could have	**Dependencies:** FR-14, FR-18	**Conflicts:** None
Supporting Materials: None		
History: Created October 16, 2026 by Jash Jivani.		

Requirement #: FR-23	**Requirement Type:** 9	**Event/BUC/PUC #:** BUC 1, 2 / PUC-2
Description: The product shall allow the owner to upload one image (JPG or PNG, up to 5 MB) for each product.		
Rationale: Pictures help the owner recognize pieces and prepare for the website.		
Originator: Owner of Lylac Jewels (client)		
Fit Criterion: An uploaded image appears on the product's page; a file of another type or larger than 5 MB is refused.		
Customer Satisfaction: 3	**Customer Dissatisfaction:** 1	
Priority: Could have	**Dependencies:** FR-4	**Conflicts:** None
Supporting Materials: None		
History: Created October 16, 2026 by Jash Jivani.		

Non-functional Requirements

10. Look and Feel Requirements

10a. Appearance Requirements

Requirement type 10 means a look and feel requirement.

Requirement #: LF-1	**Requirement Type:** 10a	**Event/BUC/PUC #:** All PUCs
Description: The product shall use the Lylac Jewels brand colors (lilac, plum and rose gold, on a white background) throughout the administrative pages.		
Rationale: The owner wants the tool to feel like part of her brand and to match the storefront website.		
Originator: Owner of Lylac Jewels (client)		
Fit Criterion: All pages use only the agreed brand color palette (no more than 5 colors plus black, white and greys), and the owner confirms the colors match the brand at the design review.		
Customer Satisfaction: 4	**Customer Dissatisfaction:** 2	
Priority: Should have	**Dependencies:** None	**Conflicts:** None
Supporting Materials: None		
History: Created October 16, 2026 by Jash Jivani.		

Requirement #: LF-2	**Requirement Type:** 10a	**Event/BUC/PUC #:** PUC-1, PUC-8
Description: The product shall display the Lylac Jewels name or logo on the sign-in page and in the header of every page.		
Rationale: Identifies the system and reinforces the brand.		
Originator: Owner of Lylac Jewels (client)		
Fit Criterion: The logo or name is visible on the sign-in page and on every administrative page.		
Customer Satisfaction: 3	**Customer Dissatisfaction:** 2	
Priority: Should have	**Dependencies:** None	**Conflicts:** None
Supporting Materials: None		
History: Created October 16, 2026 by Jash Jivani.		

Requirement #: LF-3	**Requirement Type:** 10a	**Event/BUC/PUC #:** All PUCs
Description: The product shall have an uncluttered appearance, with the same header, navigation menu position and button styles on every page.		
Rationale: A busy owner should always know where to find each task (Goal G4).		
Originator: Owner of Lylac Jewels (client)		
Fit Criterion: On review of all pages, the navigation is in the same place with the same items, and no page shows more than one primary (highlighted) action button.		
Customer Satisfaction: 4	**Customer Dissatisfaction:** 3	
Priority: Must have	**Dependencies:** None	**Conflicts:** None
Supporting Materials: None		
History: Created October 16, 2026 by Jash Jivani.		

Requirement #: LF-4	**Requirement Type:** 10a	**Event/BUC/PUC #:** PUC-3, 5, 6, 7, 8
Description: The product shall show order statuses and product availability as labeled badges that use both text and color.		
Rationale: Statuses must be recognized quickly and must not depend on color alone.		
Originator: Jash Jivani (developer)		
Fit Criterion: All five order statuses and both availability states have a distinct badge, and each can be identified correctly in a greyscale screenshot.		
Customer Satisfaction: 4	**Customer Dissatisfaction:** 3	
Priority: Must have	**Dependencies:** None	**Conflicts:** None
Supporting Materials: None		
History: Created October 16, 2026 by Jash Jivani.		

Requirement #: LF-5	**Requirement Type:** 10a	**Event/BUC/PUC #:** All PUCs
Description: The product shall be displayed without clipped or overlapping content on laptop screens from 1280 × 720 pixels upward.		
Rationale: The owner works on a laptop (Section 3e).		
Originator: Jash Jivani (developer)		
Fit Criterion: Every page viewed at 1280 × 720 and 1920 × 1080 at 100% zoom shows all labels and buttons fully, with no horizontal scrolling.		
Customer Satisfaction: 3	**Customer Dissatisfaction:** 4	
Priority: Must have	**Dependencies:** None	**Conflicts:** None
Supporting Materials: None		
History: Created October 16, 2026 by Jash Jivani.		

10b. Style Requirements

Requirement #: LF-6	**Requirement Type:** 10b	**Event/BUC/PUC #:** All PUCs
Description: The product shall appear elegant and professional, in keeping with a boutique jewelry brand.		
Rationale: The owner will use the tool every day and wants it to reflect the quality of her brand.		
Originator: Owner of Lylac Jewels (client)		
Fit Criterion: At the design review and the final demonstration, the owner rates the statement “The system looks elegant and professional, like Lylac Jewels” at 4 or higher on a 5-point scale.		
Customer Satisfaction: 4	**Customer Dissatisfaction:** 2	
Priority: Should have	**Dependencies:** None	**Conflicts:** None
Supporting Materials: None		
History: Created October 16, 2026 by Jash Jivani.		

Requirement #: LF-7	**Requirement Type:** 10b	**Event/BUC/PUC #:** All PUCs
Description: The product shall use no more than two font families, with clearly larger headings than body text.		
Rationale: A consistent, readable typographic style looks polished and is easy to scan.		
Originator: Jash Jivani (developer)		
Fit Criterion: A review of the style sheet shows at most two font families, and on every page the page heading is larger than all other text.		
Customer Satisfaction: 3	**Customer Dissatisfaction:** 2	
Priority: Should have	**Dependencies:** None	**Conflicts:** None
Supporting Materials: None		
History: Created October 16, 2026 by Jash Jivani.		

Requirement #: LF-8	**Requirement Type:** 10b	**Event/BUC/PUC #:** All PUCs
Description: The product shall use friendly, plain business words that the owner uses (as defined in the glossary) and no programming or database terms on screen.		
Rationale: The owner is not technical; familiar words make the product feel approachable (Goal G4).		
Originator: Owner of Lylac Jewels (client)		
Fit Criterion: A review of all page text and messages finds no technical terms such as “record ID,” “null,” “SQL” or “error 500,” and every action label matches a glossary term.		
Customer Satisfaction: 4	**Customer Dissatisfaction:** 3	
Priority: Must have	**Dependencies:** None	**Conflicts:** None
Supporting Materials: Section 4a Glossary		
History: Created October 16, 2026 by Jash Jivani.		